
A Student's Proof Pamphlet

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Preface

The first noticeable fact about mathematics is that it deals with abstract ideas, not tangible objects. For a preliminary illustration of this point, it suffices to consider a subject with which we are all familiar: arithmetic. There we perform operations on numbers without worrying about how to relate them in each case to concrete objects. The number five can of course appear in very concrete settings: there can be five sheep in a field, five books on a shelf, or five players on a team. But five itself is not a tangible object. We cannot see or touch it any more than we can see or touch time. It is an abstraction, a property shared by all collections containing as many elements as there are fingers on a hand. And of all such collections it is true that, if we add one more element, we get a collection of six. And if we add one more, we obtain a collection of seven. From this simple process of adding ones we can construct arbitrarily large numbers: a number larger than the number of people on Earth, larger than the number of grains of sand on Mars, larger than the number of atoms in the universe. That in practice counting is limited is not relevant. An abstraction is made from the process, and from this abstraction the notion of infinity enters quite naturally.

Abstractions of this sort are characteristic of the whole of mathematics, which is to say that mathematical objects are *imaginary*. Sculptors have marble and carpenters have wood, but mathematicians have no material from which to carve their theorems. Their imagination must supply both the stone and the chisel, and logic is the art of applying the one to the other.

It is not, however, an art that school mathematics does much to teach. There, mathematics is for the most part presented as a set of procedures: a formula for the area of a circle, a method for solving a quadratic equation, a rule for differentiating a product. The student who learns them and applies them correctly is rewarded, and the question of why they hold is seldom asked. At university this changes. The “didactical contract” is rewritten. The correct answer, which until then was the whole of the task, becomes only a part of it; the student must

now know why it is correct, and be able to show it. This is no loss. Behind the new demand lies a larger and richer mathematics than the one left behind. But it is a demand for which many students are unprepared, and many feel like foreigners and get lost.

The texts that might help them are, broadly, of two kinds. The first sets aside a few pages on the methods of proof at the opening of a course in calculus or in linear algebra. This is better than silence, but it is slight. The methods of proof ask for the same patient attention we give to any other part of mathematics, and a few pages cannot supply it. The second is the coursebook devoted entirely to proof. This is generous; for the present purpose, perhaps too generous. Such books tend to bring with them a considerable apparatus (sets and relations, functions and quantifiers, the formal machinery of logic), much of which the beginner does not yet need in order to follow a simple proof, still less to find one.

This pamphlet is meant to fill the space between them. It treats the methods of proof at greater length than an introductory chapter permits and with fewer prerequisites than a dedicated coursebook demands. Its examples are taken from mathematics with which the reader is already familiar, so that attention may rest on the reasoning and not on the objects reasoned about. Some of what is proved will seem elementary, perhaps crossing the border into the obvious. That is intended. The aim is not to teach new mathematics, but to try to spur some curiosity, give a little taste for mental work, and provide some means of doing it.

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Preamble

All the proofs we will see in this book will involve integers, so it could be useful for us to recall some simple facts about them, if only to agree on some notation and terminology. The **integers** are the numbers

$$\dots, -2, -1, 0, 1, 2, \dots$$

and the set of all of them is denoted $\mathbb{Z}$. The integers are **closed** under addition, subtraction, and multiplication, but not under division. This means that if m and n are integers, then so are $m + n$, $m - n$, and $m \cdot n$, but m/n need not be: 1 and 2 are integers, but $1/2$ is not.

The integers fall into two classes, the **even** and the **odd**. An integer is even if it can be written as $2k$ for some integer k . Thus 6 is even, since $6 = 2 \cdot 3$; so is 0 since $0 = 2 \cdot 0$; and so is -4 since $-4 = 2 \cdot (-2)$. An integer is odd if it can be written as $2k + 1$ for some integer k . Thus 7 is odd, since $7 = 2 \cdot 3 + 1$, and so is -3 since $-3 = 2 \cdot (-2) + 1$. Every integer is either even or odd, and none is both.

Evenness is a special case of a more general notion. We say that an integer m divides n if $n = mk$ for some integer k . The same relation is also expressed by saying that m is a divisor of n , that n is a multiple of m , or that n is divisible by m . When m divides n , we write $m|n$, and when it does not we write $m \nmid n$. For example, $3|12$ since $12 = 3 \cdot 4$. On the other hand, $5 \nmid 12$: the multiples of 5 nearest to 12 are 10 and 15, and 12 is neither. Note that $m|n$ is something that is either true or false, while m/n is a number. When m is not zero, $m|n$ says exactly that n/m is an integer: $12/3 = 4$, but $12/5 = 2.4$. In this language, an integer is even precisely when 2 divides it.

Every integer n is divisible by 1 and itself, since $n = 1 \cdot n$. Some positive integers have other positive divisors as well, and some do not. Six does: $6 = 2 \cdot 3$. Seven does not, for its only positive divisors are 1 and 7. An integer greater than 1 is called **prime** if its only positive divisors are 1 and itself, and **composite** otherwise. Thus 7 is prime and 6 is composite. The number 1 is neither. The first ten primes are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29.$$

Every integer greater than 1 is either prime or can be written as a product of primes. For example, $12 = 2 \cdot 2 \cdot 3$ and $30 = 2 \cdot 3 \cdot 5$.

The integers are ordered:

$$\dots < -3 < -2 < -1 < 0 < 1 < 2 < 3 < \dots$$

The ordering has no upper bound and no lower bound: there is no largest integer and no smallest. Fix your attention now on those integers greater than 0, and choose among them as you please one, a handful, or a great many, provided only that you choose at least one. Among those you have chosen there is always a smallest. This is known as the **well-ordering principle**. The restriction to positive integers is essential: among the negative integers, there is no smallest, but there is a largest. On occasion it will be useful to apply the principle to the nonnegative integers, that is, to include 0 among those we choose from.

The **absolute value** of an integer n , written $|n|$, is n itself if $n \geq 0$ and $-n$ if $n < 0$. Thus $|5| = 5$, $|-3| = 3$ and $|0| = 0$. It measures how far n lies from 0, without regard to direction, so it is never negative.

Finally, sums of many terms occur often enough to deserve a notation of their own. We write

$$a_1 + a_2 + \dots + a_n = \sum_{i=1}^n a_i,$$

and read this as “the sum of a_i from $i = 1$ to n .” The letter i is called the index. It takes each integer value from 1 to n in turn, and the terms so obtained are added. Thus

$$\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30.$$

The index need not start at 1:

$$\sum_{i=0}^3 2^i = 2^0 + 2^1 + 2^2 + 2^3 = 1 + 2 + 4 + 8 = 15.$$

Nor does its name matter, so

$$\sum_{i=1}^n a_i = \sum_{k=1}^n a_k.$$

A few manipulations follow directly from the meaning of the notation. For example, a factor common to every term may be taken outside the sum or brought inside it:

$$c \sum a_i = \sum c a_i,$$

and two sums over the same range of the index may be added term by term

$$\sum a_i + \sum b_i = \sum (a_i + b_i).$$

The index may also be shifted. For example, writing $k = i + 1$, as i runs from 0 to $n - 1$, k runs from 1 to n , so that

$$\sum_{i=0}^{n-1} a_{i+1} = \sum_{k=1}^n a_k.$$

The same sum written in two ways.

CHAPTER 1

Logic

1.1. Deduction

We draw conclusions constantly, and for the most part without noticing that we do so. An old story shows the process at work, and shows too how quickly it can run.

A bishop was once the guest of honor at a country party given by a duke. At one point the duke left his guests to order something from his servants, and the bishop, eager to entertain the remaining company, told them that long ago, when he was a newly ordained priest, the first person whose confession he heard was a multiple murderer of the vilest kind. The duke thereupon returned, clapped the bishop on the shoulder and said, “The bishop and I are very old acquaintances. In fact, I was the first person whose confession he heard.” The rest of the guests hastily left.

No one present had seen the duke commit any crime. No accusation had been made; the bishop had named nobody and the duke had confessed to nothing but an old acquaintance. Yet the guests were in no doubt, and their haste to be gone shows it. The two things they were granted – that the first confession the bishop heard was a murderer’s and that the duke was his first penitent – settled the matter by themselves. The conclusion was already contained in them, and the guests had only to put them together.

The same putting-together happens in humbler cases, though without the scandal that fixes this one to memory. All men are mortal; Socrates is a man; therefore Socrates is mortal. Think of all mortals as one large company. The first statement says that every man belongs to it. The second says that Socrates is a man. Then Socrates, like every other man, belongs to the company of mortals, which is only to say that he is mortal.

This is **deduction**: the passing from statements we grant or assume to another which they compel. The statements we start from are called the **premises** and the one we arrive at is the **conclusion**. In neither of our examples did we need to consult the world. We did not have to

witness the duke's crimes or wait for Socrates's death. We needed only to accept the premises, and the conclusion was inescapable.

Notice that in so doing we never asked whether the premises were actually true. That is a separate question, and often no concern of logic at all. There is a distinction between *correct information* and *correct reasoning*. Ideally we want both our reasoning and our information to be correct, but a deductive argument tells us only what we are bound to accept once the premises are granted. A single such step, from premises to conclusion, is an **inference**.

1.2. Statements

The premises of an argument, and its conclusion, are **statements**. A statement is a sentence that is either true or false – the one or the other, and not both. "The duke is a murderer" is a statement: though we may not know which of the two it is, one of the two it must be. "Seven is larger than five" is a statement, and so is "three is larger than nine" – the former true, the latter false.

Not every sentence is of this kind. "Turn off the light!" is a command; it is a request for action, not an assertion. One cannot call "Turn off the light!" true or false. "Is the window open?" is a question, and a question asserts nothing – it asks. Only sentences that assert something, that lay down how matters stand, are true or false, and only these will concern us.

From statements we can make further statements. Given "the light is lit" and "the window is open," we may say, "the light is lit *and* the window is open," or "the light is lit *or* the window is open," or, of the first alone, "the light is *not* lit." The small words that join or alter statements in this way – **not**, **and**, **or** – are called connectives. We shall take them one at a time.

The simplest of them is *not*, which alters a single statement rather than joining two. To negate a statement is to assert that it is not so. If "the light is lit" is true, then "the light is not lit" is false; and if the first is false, the second is true. Negation reverses truth and falsehood, and does so in every case. In the standard notation, a letter such as P stands for any statement, and $\neg P$, read "not P ", for its denial. The working of each connective can be set out in what is called a truth table. For **negation** it is as follows, where T stands for "true" and F for "false":

P	$\neg P$
T	F
F	T

The column under P lists the possible truth values of P , that is, whether it is true or false. Here we have only one statement, so there are only two possibilities. The second column shows the result for $\neg P$. Since putting *not* in front of a statement reverses its truth value, doing it twice restores the original statement: $\neg(\neg P)$ is true exactly when P is. If P is “the light is lit,” then $\neg P$ is “the light is not lit,” which is to say “the light is unlit,” and so $\neg(\neg P)$ is “the light is not unlit,” which is to say “the light is lit.”

The word *and* joins two statements into one, called their **conjunction**, which is true only when both parts are true. One false part is enough to condemn the whole. This is exactly what we ordinarily mean by *and*. Letting P and Q stand for statements and $\wedge$ for *and*, there are now four possible combinations of truth values, and the truth table is as follows:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

The word *or* is less straightforward, for in ordinary speech it says two different things. When a meal comes with coffee or tea, it comes with one or the other, not both: the *or* shuts out the case where you take both. But when a post is declared open to any citizen or resident, a person is not turned away for being both; the citizen who also resides there is welcome. Here the *or* admits the case where both hold. The same small word thus sometimes excludes the both-case and sometimes includes it. Logic cannot proceed on such ambiguity, so it must settle on one meaning. Let $\vee$ stand for *or*. $P \vee Q$ is counted true when P is true, when Q is true, and when both are true; it is false only when both are false. This is the inclusive *or*, the *or* of the citizen and resident rather than that of the coffee and tea. The choice is partly a matter of convention, but some choice had to be made: a word used in logic must have its meaning fixed in advance, and this is the meaning fixed upon. The truth table is as follows:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Notice what the three words have in common. In each case the truth value of the compound statement is settled entirely by the truth values

of its parts. To judge whether "the light is lit and the window is open" is true, we need know nothing of lights or windows beyond whether each part is true; the word *and* does the rest.

There remains one connective, and it is the one we have been moving toward. Most theorems in mathematics have the form *if this, then that*: if a number is divisible by four, then it is even; if two sides of a triangle are equal, then so are two of its angles. The words that join the one to the other are *if...then*, and one might expect them to be settled as readily as *and* or *or*. They are not. Though they look as plain as the rest, they do not behave as the rest do, and they are best given a section to themselves.

1.3. Conditional Statements

The connective we set aside is called the **conditional**. A conditional statement joins two statements by the words *if... then*. For example, "if it rains, then the ground is wet." The statement following *if* is the **antecedent**; the statement following *then* is the **consequent**. Here the antecedent is "it rains" and the consequent is "the ground is wet." With P and Q standing for statements as before, we write $P \Rightarrow Q$, and read this as "if P , then Q " or " P **implies** Q ."

But what does it mean for $P \Rightarrow Q$ to be true? To settle this, we consider a conditional phrased as a promise. A professor tells a student "if you earn an A on the final exam, then I will give you an A for your final grade." In which case can we say that the professor lied, that is, that the statement was false?

There are four possibilities. Let P be "the student earns an A on the final exam" and Q be "the student receives an A for the final grade." In the first possibility, the student earns an A on the final exam and later receives an A for the final grade. This is the situation when P and Q are both true. The professor did as promised, so $P \Rightarrow Q$ is true. In the second, the student earns an A on the final exam but does not receive an A for the final grade. This is the situation when P is true and Q is false. The professor did not do as promised, so $P \Rightarrow Q$ is false. The first two rows of the truth table are therefore:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F

Next, suppose the student did not earn an A on the final exam but even so received an A as the final grade. This is the case when P is false and Q is true. Did the professor break the promise? No, for the promise said nothing about what would happen if the student did not

earn an A on the exam. $P \Rightarrow Q$ is therefore not false, and since every statement is either true or false, it is true:

P	Q	$P \Rightarrow Q$
F	T	T

Finally, suppose the student earned an A neither on the final exam nor as the final grade, so that P and Q are both false. Again the professor broke no promise, since the promise applied only if the student earned an A on the exam. So $P \Rightarrow Q$ is true:

P	Q	$P \Rightarrow Q$
F	F	T

The promise fails, then, in only one case: when the student earns an A on the final exam but not as the final grade. This is the rule we were seeking. A conditional is false when its antecedent is true and its consequent false, and true in every other case. In particular, it is true whenever the antecedent is false, for a promise says nothing about the case in which its condition never arises, and cannot be broken there. The complete truth table is as follows:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

A mathematical statement of the same form makes the same kind of promise. "If a number is divisible by six, then it is divisible by three" promises only this: every number divisible by six is also divisible by three. It says nothing about numbers that are not divisible by six. Nine is not divisible by six, though it is divisible by three; four is divisible by neither. Neither number breaks the promise, for neither is a number the promise is about. They are like the student who did not earn an A on the exam.

How, then, could the statement be false? Only if there were a number divisible by six but not by three. That is the one case that would break the promise, and to prove the statement is to show that this case never occurs.

Where would such a number be found, if it existed? Not among the numbers that are not divisible by six; we have just seen that none of them can break the promise. It could only be among the numbers that are divisible by six. So it is to these that we turn. We take a number divisible by six, any such number and not a particular one, and ask

whether it could fail to be divisible by three. If we can show that it cannot, the statement is proved.

This is what it means, in practice, to prove $P \Rightarrow Q$. We assume that P is true. This sets aside the last two rows of the table, where P is false; they could never have made the conditional false. The first two rows remain. In the first, Q is true and the conditional holds; in the second, Q is false and the conditional fails. Our task is to rule out the second row, and we do so by showing that once P is assumed, Q must follow.

Notice that the direction matters. To prove $P \Rightarrow Q$ we assume P and show Q , not the other way around. If we turn the statement around, we get a different one. The conditional $Q \Rightarrow P$ is called the **converse** of $P \Rightarrow Q$, and the two are not the same. "If a number is divisible by six, then it is divisible by three" is true. Its converse, "if a number is divisible by three, then it is divisible by six," is false: nine is divisible by three but not by six. A statement and its converse do not stand or fall together.

Sometimes, though, $P \Rightarrow Q$ and $Q \Rightarrow P$ are both true. We then say that P and Q are **equivalent**, and write $P \Leftrightarrow Q$, which is read " P if and only if Q ." This means that P and Q are true in exactly the same cases: wherever one holds, so does the other, and wherever one fails, so does the other. They need not say the same thing. A number is even if and only if its last digit is even. One side is a fact about the whole number, the other a fact about a single digit, and yet the two stand or fall together. If we are told that a number is even, we know its last digit is even; if we are told that its last digit is even, we know the number is even. To be told either is to be told both. The truth table is therefore:

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

One thing has been passed over. Take the statement "if a number is divisible by six, then it is divisible by three." What it really asserts is that *every* number divisible by six is also divisible by three. Its converse "if a number is divisible by three, then it is divisible by six" is false because it does not hold for *every* number divisible by three: nine, for example, is divisible by three but not by six. But it does hold for *at least one*: eighteen is divisible by both. Words such as *every* and *at least one* tell us how many of the things in question a statement is about. They are called **quantifiers**, and they are the subject of the next section.

1.4. Quantifiers

Ordinary language has many words for saying how many: *all, every, any, some, none, at least one*. Various as they are, they do the work of just two. One says that something holds of everything in question: *all* politicians are dishonest; *every* number divisible by six is divisible by three. The other says that it holds of at least one thing: *some* politician is dishonest; *at least one* number divisible by three is divisible by six. The first is called the **universal quantifier**, the second the **existential quantifier**.

Quantifiers matter most when we come to negate statements, for here ordinary speech easily goes wrong. What does it take to show that "all politicians are dishonest" is false? Not that all politicians are honest; one honest politician is enough. And what does it take to show that "at least one politician is dishonest" is false? Here one honest politician is not enough; every politician must be honest. Let U stand for the universal statement and E for the existential one:

- U : All politicians are dishonest,
- E : At least one politician is dishonest,
- $\neg U$: At least one politician is honest,
- $\neg E$: All politicians are honest.

We see that the negation turns the one quantifier into the other: the negation of a universal statement is existential, and the negation of an existential statement is universal. This has a practical consequence. To show that a universal statement is false, we need only show that its negation is true, and its negation is existential: it asks for a single instance. One honest politician refutes the claim that all politicians are dishonest. Such an instance is called a **counterexample**. We have already met one: nine, which is divisible by three but not by six, is a counterexample to the claim that every number divisible by three is divisible by six. There is an imbalance here worth remembering. To prove a universal statement, we must show that it holds in every case; to disprove it, one case is enough. The same holds in reverse. To prove an existential statement, a single instance is enough: eighteen, being divisible by both three and six, shows that some number divisible by three is divisible by six. But to show that an existential statement is false, we must show that its negation is true, and its negation is universal. We must show that no instance exists anywhere.

1.5. The Conditional Revisited

Recall from Section 1.3 that $P \Rightarrow Q$ is false in only one case: when P is true and Q is false. Put another way, $P \Rightarrow Q$ is false exactly when $P \wedge \neg Q$ is true, and so it is true exactly when $P \wedge \neg Q$ is false. This gives us the following result, which we shall call **the implication law**:

$$P \Rightarrow Q \text{ if and only if } \neg(P \wedge \neg Q).$$

The law puts into symbols what we saw with the professor's promise: to assert $P \Rightarrow Q$ is to rule out the one case in which P is true and Q is false.

Now, suppose $A \Rightarrow B$ is true. By the implication law this means that $\neg(A \wedge \neg B)$ is true. A conjunction is true only when both its parts are true, so the order of the parts does not matter, and $\neg(\neg B \wedge A)$ is true as well. Since a double negation restores the original truth value, A may be written as $\neg(\neg A)$, which gives $\neg(\neg B \wedge \neg(\neg A))$. This is exactly the form of the implication law with $\neg B$ in place of P and $\neg A$ in place of Q . Reading it back through the law, we get $\neg B \Rightarrow \neg A$. Each step can be run in the other direction, so we have shown that

$$A \Rightarrow B \text{ if and only if } \neg B \Rightarrow \neg A.$$

The statement $\neg B \Rightarrow \neg A$ is called the **contrapositive** of $A \Rightarrow B$. The result tells us that to prove $A \Rightarrow B$ we can instead prove $\neg B \Rightarrow \neg A$, which, in fact, is the proof method we will meet in Section 2.2.

A simple example is "if someone is in Oslo, then they are in Norway." The one case this statement rules out is a person who is in Oslo but not in Norway. Its contrapositive, "if someone is not in Norway, then they are not in Oslo," rules out exactly the same case: a person who is not in Norway and yet is in Oslo. The two statements forbid the same thing, and that is why they stand or fall together. They differ only in where they begin. The original starts from Oslo and asks whether the person must be in Norway; the contrapositive starts outside Norway and asks whether the person could be in Oslo. Answer either question, and the other is answered too.

1.6. Epilogue

All of mathematics takes the form of deduction. In proving a mathematical statement, or even in solving a simple equation, we take information that is known or assumed and apply deductive logic to produce new pieces of information, but we do not ask whether the information we start with is true. It is the form, and not the content, that constitutes the proof. Put differently, we are not in the least concerned with the

truth or otherwise of our assumptions, but merely with the correctness of our deductions.

When a statement has a proof we will call it a **theorem** or a **proposition** or a **lemma** or a **corollary**. There is no clear distinction between these terms, and their usage often depends on the taste of the writer, but as a general rule we say that a theorem is an important result that will be used over and over again; a proposition is a result that is considered less important than a theorem; a lemma is typically a smaller result whose main purpose is to help us prove a proposition or a theorem; and a corollary is a result that is an immediate consequence of the proposition or theorem that has just been proved. In any event, the important thing is that a statement that carries any one of these names has a proof. A rose by any other name would smell as sweet.

Most mathematical theorems can be formulated as an implication $P \Rightarrow Q$. To prove one is to rule out the single case that would make it false: that P is true and Q is false. The most direct way to do this is to assume P and show, through a series of inferences, that Q follows. We may think of P as what we know and Q as what we want to know, and the proof is the passage from the one to the other. To prove an equivalence $P \Leftrightarrow Q$ is then to prove two implications, $P \Rightarrow Q$ and $Q \Rightarrow P$, and so to make the passage both ways.

The question now is how we can build such passages. What steps are permitted? What shapes can a proof take? Section 1.5 has given us a taste. The next chapter will give us more.

CHAPTER 2

Proof Techniques

2.1. Direct Proof

As we saw at the end of Chapter 1, to prove $P \Rightarrow Q$ we assume P and show, through a series of inferences, that this leads to Q . In this setting P is called the hypothesis and Q the conclusion, and a proof that passes directly from the one to the other is called a direct proof. Each step in it must be licensed, either by a definition or by a fact already agreed.

Where should the steps begin? Almost always with the definitions. The first statement we shall prove, that the sum of two even integers is even, is a conditional in disguise, for it says that *if* two integers are even, *then* their sum is even. But the word even, taken by itself, offers nothing to work with. Its definition does, for it tells us that each of the two integers can be written as $2k$ for some integer k . In this form the hypothesis can be handled by ordinary arithmetic, and the arithmetic, with luck, delivers the form named in the conclusion.

Proposition 2.1.1

The sum of two even integers is even.

Proof. Let x and y be even integers. By the definition of an even number, $x = 2m$ for some integer m , and $y = 2n$ for some integer n . Then,

$$x + y = 2m + 2n = 2(m + n).$$

Now, $m + n$ is an integer, since the sum of two integers is again an integer. Thus, $x + y$ is twice an integer, which is to say it is even. $\square$

Observe what the last step did. We factored the 2, not to simplify the arithmetic, since $2m + 2n$ was simple enough, but to make the form of an even number visible in the result. The proof begins by writing the definition in and ends by reading the definition out. Observe, too, the letters. We did not check that $2+4$ and $6+8$ are even and grow more confident with each case. Such checks would no more prove the statement than measuring a thousand right-angled triangles would prove the theorem of Pythagoras. Since m and n can be any integers, the argument

covers every pair of even integers at once. That is why proofs are written in letters where calculation would use numbers. The small square at the end of the proof is called a **halmos** and marks that the proof is complete. It stands in for QED, *quod erat demonstrandum*, “which was to be demonstrated.”

The next proof has the same shape as the first, but its last step asks a little more of us. In the first proof, once the 2 was factored out, the form of an even number $2(m + n)$, lay ready on the page. Here the result must be coaxed into the form of an odd number $2k + 1$ before the conclusion can be recognized. We shall have occasion to use this statement again in a later section.

Proposition 2.1.2

If n is odd, then n^2 is odd.

Proof. Suppose n is odd. Then, $n = 2k + 1$ for some integer k , and so

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 has the form $2m + 1$ where $m = 2k^2 + 2k$, and is therefore odd. $\square$

The next proof carries the method away from even and odd numbers, where our first two proofs kept it, and applies it to divisibility. It also has two hypotheses rather than one, and its central step is to substitute the one into the other. The method itself is unchanged, for it does not depend on the subject. Whatever the definitions, we write the hypotheses in the form they supply, work forward by ordinary steps, and recognize the conclusion when it appears.

Proposition 2.1.3

Let a , b and c be integers. If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof. Assume that $a \mid b$ and $b \mid c$. Writing each hypothesis in the form given by the definition of divisibility, we have $b = am$ and $c = bn$ for some integers m and n . Inserting the former into the latter,

$$c = bn = (am)n = a(mn).$$

Since mn is an integer, c is a times an integer, and so $a \mid c$. $\square$

For our last example the expression cannot be handled in one piece; it must first be separated into cases.

Proposition 2.1.4

Let n be a positive integer. Then, $1 + (-1)^n(2n - 1)$ is a multiple of 4.

Before proving this, let us see where the difficulty lies. The expression contains the factor $(-1)^n$, whose value is not fixed: it is 1 when n is even and -1 when n is odd. Until we know which, we cannot simplify the expression, for the two cases lead to different arithmetic. But every integer is either even or odd, and none is both. So we take the two cases in turn and prove the claim in each. Since no third case remains, the claim holds for every n .

Proof. We divide into two cases according to whether n is even or odd. Suppose first that n is even. Then, $(-1)^n = 1$, and

$$1 + (-1)^n(2n - 1) = 1 + (2n - 1) = 2n.$$

Since n is even, $n = 2k$ for some integer k , and so the expression equals $2(2k) = 4k$, which is a multiple of 4.

Suppose now that n is odd. Then, $(-1)^n = -1$, and

$$1 + (-1)^n(2n - 1) = 1 - (2n - 1) = 2 - 2n.$$

Since n is odd, $n = 2k + 1$ for some integer k , and so $2 - 2n = 2 - 2(2k + 1) = -4k = 4(-k)$, again a multiple of 4. $\square$

2.2. Proof by Contraposition

In every proof of the last section, the definitions did the heavy work. We should not expect them always to.

Consider the statement “if n^2 is even, then n is even”. Let us try to prove it as we proved the others. We assume the hypothesis and write it in its defining form: $n^2 = 2k$ for some integer k . We are to reach the conclusion that $n = 2m$ for some integer m . But from $n^2 = 2k$ there is nowhere to go. The equation tells us what n^2 is, and the conclusion asks about n ; and no rule of arithmetic carries us from the one to the other. We could write $n = \sqrt{2k}$, but nothing in this expression shows that n is an integer, let alone an even one. The direct method stalls at its first step.

It would be different if the roles were exchanged, so that we assumed something about n and had to conclude something about n^2 . Then, the direct method would have something to work on, for from n we can always form n^2 . Section 1.5 shows how to make the exchange. A conditional $P \Rightarrow Q$ is equivalent to its contrapositive $\neg Q \Rightarrow \neg P$, so

to prove “if n^2 is even, then n is even” we may instead prove “if n is not even, then n^2 is not even,” that is, “if n is odd, then n^2 is odd.” Proving a conditional by proving its contrapositive is called **proof by contraposition**.

Proposition 2.2.1

Let n be an integer. If n^2 is even, then n is even.

Proof. We prove the contrapositive: if n is odd, then n^2 is odd. But this is exactly what we proved in Proposition 2.1.2. The contrapositive therefore holds, and with it the statement itself. $\square$

This first example shows the method at its plainest. The swap alone did the work: the statement we could not begin with was equivalent to one we had already finished. Nothing remained but to notice it. The next example is less fortunate. Again the original statement gives us nothing to begin with, and again its contrapositive does, but this time the contrapositive is not a result we already have, and we must prove it ourselves. Nothing new is needed for this. We write the hypothesis in its defining form, carry out the arithmetic, and recognize the conclusion: a direct proof from beginning to end. Only its subject, the contrapositive rather than the original, is chosen with care.

Proposition 2.2.2

If $3n + 2$ is odd, then n is odd.

Proof. We prove the contrapositive: if n is even, then $3n + 2$ is even. Suppose, then, that n is even. Then, $n = 2k$ for some integer k , and

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

Since $3k + 1$ is an integer, $3n + 2$ is even. This proves the contrapositive, and therefore the original statement. $\square$

For the last example, the statement is compound, and forming its contrapositive asks for more attention. The proof itself will be routine; the care lies in stating what is to be proved. The conclusion is “ a is even or b is even,” and we need its negation. When is this conclusion false? An *or* statement is true as soon as one of its parts is true, so it is false only when both parts are false: when a is odd and b is odd. The negation of “ a is even or b is even” is therefore not “ a is odd or b is odd,” but “ a is odd and b is odd.” If we negate carelessly, we will set

out to prove the wrong thing. Forming the contrapositive is a logical operation, and the rules of Chapter 1 govern it.

Proposition 2.2.3

Let a and b be integers. If ab is even, then a is even or b is even.

Proof. We prove the contrapositive: if a is odd and b is odd, then ab is odd. Suppose, then, that a and b are both odd. Then, $a = 2m + 1$ and $b = 2n + 1$ for some integers m and n , and

$$ab = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n$ is an integer, ab is odd. This proves the contrapositive, and therefore the original statement. $\square$

Let us be clear about what we have done. We have not introduced a new way of reasoning. Every proof in this section, once its opening line was written, proceeded as those of Section 2.1 did, by assuming a hypothesis and working forward to a conclusion. What is new is a choice made before the work begins: to prove the contrapositive in place of the given conditional. We make this choice when the hypothesis, written in its defining form, gives us nothing to calculate with, while the negated conclusion does. Nothing is supposed false, and nothing is driven to an absurdity. This is worth remembering, because the method of the next section does both, and is easily mistaken for contraposition.

2.3. Proof by Contradiction

In a direct proof of $P \Rightarrow Q$, we assume P and work forward to Q . In a proof by contraposition we assume $\neg Q$ and work forward to $\neg P$. There is a third way. Recall once more that $P \Rightarrow Q$ fails in exactly one circumstance: when P is true and Q is false. Suppose, then, that we assume P and $\neg Q$, and that this leads us to conclude that some statement C and its negation $\neg C$ are both true. A statement cannot be both true and false at once, so $C \wedge \neg C$ is always false. It is a **contradiction**. We have therefore shown that the implication $(P \wedge \neg Q) \Rightarrow (C \wedge \neg C)$ is true while its conclusion is false. But a true implication with a false conclusion must have a false hypothesis, as the last row of its truth table shows. Hence, $P \wedge \neg Q$ is false, so $\neg(P \wedge \neg Q)$ is true, and by the implication law, so is $P \Rightarrow Q$.

In short: to prove $P \Rightarrow Q$, we may assume P and $\neg Q$ and show that this leads to a contradiction, such as " $0 = 1$ " or " n is even and odd" or " $x \neq y$, yet $(x - y)^2 \leq 0$ ". Often one half of the contradiction is a fact

we already know: when we reach $0 = 1$, the other half is $0 \neq 1$. This is proof by contradiction.

Before the examples, two remarks. The first concerns the difference from contraposition, which the last section warned is easily missed. Contraposition assumes $\neg Q$, and $\neg Q$ alone; its target is fixed: it must reach $\neg P$. Contradiction assumes both $\neg Q$ and P , and its target is any contradiction at all. That openness is the strength of the method, and also its hazard.

This brings us to the second remark. In a direct proof, a false step usually leaves us stranded: the conclusion stays out of reach, and the failure warns us that something is wrong. In a proof by contradiction, there is no such warning, since we cannot always say beforehand which contradiction we are looking for, and a false step is perfectly capable of producing one. The method therefore asks for more vigilance, not less.

A number is rational if it can be written as a fraction p/q of two integers, with $q \neq 0$. Every integer is rational, since $n = n/1$. We know that among the positive integers there is a smallest, namely 1, and one might expect that among the positive rationals there would be a smallest too. There is not.

Proposition 2.3.1

There is no smallest positive rational number.

Proof. Assume for the sake of a contradiction that there is a smallest positive rational number, call it r . Since r is positive, so is $r/2$, and since r is rational, so is $r/2$. But $r/2 < r$, so $r/2$ is then a positive rational number smaller than r , contradicting that r was the smallest. Hence, no smallest positive rational number exists. $\square$

This is the kind of statement contradiction is made for: a claim that something does not exist. We supposed that it did, and the supposition undid itself by producing a rational smaller than the smallest. Notice that the assumption did real work: without an r to halve, there would be no argument at all. Our next claim has the same shape and is far more famous. It says that $\sqrt{2}$ cannot be written as a fraction, that is, that there are no integers p and q with $(p/q)^2 = 2$.

Proposition 2.3.2

$\sqrt{2}$ is irrational.

Proof. Assume for the sake of a contradiction that $\sqrt{2}$ is rational, so that

$$\sqrt{2} = \frac{p}{q}$$

for some integers p and q with $q \neq 0$. We may assume the fraction is in lowest terms – that p and q have no common factor greater than 1 – for if they had one, we could divide it out and obtain a fraction of the same value with smaller terms. Squaring both sides gives $2 = p^2/q^2$, and hence

$$p^2 = 2q^2.$$

This means that p^2 is even, and, by Proposition 2.2.1 so is p . Write $p = 2k$ where k is some integer. Then, $(2k)^2 = 2q^2$, that is, $4k^2 = 2q^2$, and hence,

$$q^2 = 2k^2.$$

This means that q^2 is even, and by Proposition 2.2.1 again, q is even. But now p and q are both even, and so both are divisible by 2, contradicting our assumption that they have no common factor. Therefore, $\sqrt{2}$ is irrational. $\square$

One of the most famous proofs in the history of mathematics is Euclid's proof that there are infinitely many primes, and a book on proof that left it out would be negligent. It, too, has the shape this method is made for. To say that the primes are infinite is to say that there is no last prime, and so we suppose that there is one, and that the primes can therefore all be listed.

Proposition 2.3.3

There are infinitely many primes.

Proof. Assume, for the sake of a contradiction, that there are only finitely many primes, say n in all, and list them in increasing order as $p_1, p_2, \dots, p_n$, so that p_n is the last. Let

$$N = p_1 p_2 \cdots p_n$$

be their product, and consider $N + 1$.

No prime on the list divides $N + 1$. For take any p_i . Since it is one of the factors of N , we have $N = b p_i$ for some integer b . If p_i also divided $N + 1$, say $N + 1 = a p_i$, then

$$(N + 1) - N = a p_i - b p_i = p_i(a - b) = 1,$$

and p_i would divide 1, which no prime does.

But since $N + 1$ is greater than 1, some prime divides it: $N + 1$ itself if it is prime, and one of its prime factors if it is not. This prime cannot

be on our list, since no prime on the list divides $N + 1$. So there is a prime missing from the list, contradicting our assumption that the list contains every prime. Hence there are infinitely many primes. $\square$

Each of these proofs supposed the very thing the theorem forbids – a smallest positive rational, a fraction squaring to 2, a finite list of primes – and followed the supposition until it collapsed. This is what contradiction does that the other methods often cannot: when a theorem denies that something exists, there is usually nothing to build toward, and the natural course is to assume the thing exists and watch it undo itself. One assumption is made, and one impossibility is reached, and between them the theorem is proved.

2.4. Mathematical Induction

Sometimes we want to prove that a statement about a positive integer n , which we write $P(n)$, holds for every n . We cannot check the cases one at a time, for they never end. Instead we use a method called **mathematical induction**, which has two steps:

- (1) Show that $P(1)$ is true.
- (2) Show that for every positive integer k , $P(k) \Rightarrow P(k + 1)$.

The first is called **the base case** and the second **the inductive step**. If both are established, the rest follows. The first step gives us $P(1)$. The second step applied to $P(1)$, gives us $P(2)$; applied to $P(2)$, it gives us $P(3)$; applied to $P(3)$, $P(4)$; and so on. Every positive integer is reached in its turn. Induction works like an infinite line of dominoes: to know that all of them fall, we need only know that the first one falls, and that each falling domino knocks over the next.

The domino picture persuades, but we can also prove that induction works, using the well-ordering principle and the method of the last section. Suppose $P(1)$ and $P(k) \Rightarrow P(k + 1)$ are both true, and assume, for a contradiction, that $P(n)$ is false for some positive integer n . Then there is a positive integer for which P fails, and so, by the well-ordering principle, there is a smallest one; call it m . Now, m cannot be 1, since $P(1)$ is true; so $m > 1$. Then $m - 1$ is a positive integer smaller than m , and so $P(m - 1)$ is true. But the inductive step, with $k = m - 1$, gives $P(m - 1) \Rightarrow P(m)$, and so $P(m)$ is true, contradicting the choice of m . The assumption was therefore false, and $P(n)$ holds for every positive integer n .

Proposition 2.4.1

For all positive integers n ,

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Proof. We argue by induction on n .

Base Case: If $n = 1$, the left side is just 1, and the right side is $1 \cdot 2 / 2 = 1$. The two agree, so the statement holds for $n = 1$.

Induction Hypothesis: Assume that the formula is true for some arbitrary positive integer k . That is, assume

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}.$$

Inductive Step: We must show that the formula holds for $k + 1$. That is, that

$$1 + 2 + 3 + \dots + k + (k + 1) = \frac{(k + 1)(k + 2)}{2}.$$

Begin with the left-hand side and set apart its final term

$$1 + 2 + 3 + \dots + k + (k + 1) = (1 + 2 + 3 + \dots + k) + (k + 1)$$

The sum in the first parentheses on the right side is exactly the left side of the inductive hypothesis, so we may replace it:

$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This is the right side for $k + 1$. So whenever the statement holds for k , it holds for $k + 1$. By the principle of mathematical induction, the formula holds for every positive integer n . $\square$

The whole of the inductive step turned on one move: separating the last term, so that the sum of the first k terms appeared and could be replaced by what $P(k)$ says it equals. The move itself belongs to sums, but the pattern belongs to every induction. The step never proves the case $k + 1$ from nothing; it proves it from the case k , and the work lies in making the case k appear.

The next example is an inequality, and its inductive step asks something the last one did not. With an equation, we may rewrite either side freely and the equality is never in danger. An inequality must be carried across the step, and it survives only if each operation preserves its direction. Multiplying both sides by a positive number does; multiplying by a negative number reverses it. In the proof below we shall

multiply by $1 + x$, and this will be safe only because the hypothesis $x > -1$ ensures that $1 + x$ is not negative.

Proposition 2.4.2: Bernoulli's Inequality

Let $x > -1$. For all positive integers n , $(1 + x)^n \geq 1 + nx$.

Proof. Fix $x > -1$, and argue by induction on n .

Base Case: For $n = 1$, the left side is just $1 + x$, and the right side is $1 + x$. They are equal, so the inequality holds.

Induction Hypothesis: Assume that the inequality holds for some positive integer k :

$$(1 + x)^k \geq 1 + kx$$

Inductive Step: We must show that the inequality holds for $k + 1$. That is, that

$$(1 + x)^{k+1} \geq 1 + (1 + k)x.$$

Write the left side as $(1 + x)^{k+1} = (1 + x)^k (1 + x)$. From the induction hypothesis, we have that $(1 + x)^k \geq 1 + kx$. Since $1 + x > 0$, we get

$$(1 + x)^{k+1} \geq (1 + kx)(1 + x).$$

Expanding the right side we get

$$(1 + kx)(1 + x) = 1 + (1 + k)x + kx^2.$$

Since $kx^2 \geq 0$, subtracting it from the right side will only make that side smaller, so

$$(1 + kx)(1 + x) \geq 1 + (1 + k)x.$$

Combining the two inequalities,

$$(1 + x)^{k+1} \geq 1 + (1 + k)x,$$

as desired. By the principle of mathematical induction, the inequality holds for every positive integer n . $\square$

Here is one more, written more compactly.

Proposition 2.4.3

For every positive integer n , the number $n^3 - n$ is divisible by 3.

Proof. We argue by induction on n . For $n = 1$, we have $n^3 - n = 1 - 1 = 0$, and 0 is divisible by 3, since $0 = 3 \cdot 0$. So the statement holds for $n = 1$. Next, assume that the statement holds for some positive integer k . This

means that $k^3 - k = 3m$ for some integer m . We must show that 3 divides $(k+1)^3 - (k+1)$. Expanding this expression we get,

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k - k = (k^3 - k) + 3k^2 + 3k.$$

The induction hypothesis has appeared. Hence,

$$(k+1)^3 - (k+1) = 3m + 3k^2 + 3k = 3(m + k^2 + k).$$

Since $m + k^2 + k$ is an integer, $(k+1)^3 - (k+1)$ is three times an integer, and so it is divisible by 3. By induction, the statement holds for all positive integers n . $\square$

2.5. On Finding a Proof

The proofs in this chapter were shown finished. Each began in the right place, chose the right method, and went straight to its end. That is how proofs are written, but it is not how they are found. The cases tried, the attempts that stalled, the wrong turns: none of it appears. A proof records the path that worked, not the ground covered in finding it, and so it looks inevitable. It is not. This is worth remembering when reading proofs in a textbook. The student who stalls on a first attempt is not doing anything wrong. They are doing what everyone does, at the stage that is usually left out. There are no rules that produce a proof, but there are habits that make one more likely.

The first is to read the statement carefully and unpack its definitions. What is the hypothesis, and what is the conclusion? What may we assume, and what must we show? Words such as even, odd, divides and rational give us nothing to calculate with; their definitions do. So we write down what we know and what we want, each in its defining form. It also helps to try a few cases. Examples are not a proof, but can help us see what the statement says, often why it is true, and sometimes that it is false. Only when we are confident of what is claimed, and of why we expect it to hold, are we ready to begin.

The second is to let the shape of the statement suggest a method. If the hypothesis gives us something to work with, we try a direct proof. If it gives us nothing, but the negation of the conclusion does, we try the contrapositive. If the statement denies that something exists, or says that there are infinitely many of something, we suppose the opposite and look for a contradiction. If it claims something for every positive integer, and the case $n+1$ seems to be built from the case n , we try induction. These are suggestions, not laws, but they can tell us where to begin.

The last is to ask whether we have met the problem, or one like it, before. What unlocks a problem is usually a bright idea, and bright ideas are the good fortune of past experience. This is why proofs are learned by reading proofs. Each one studied is added to the store from which later ideas are drawn, and the more proofs we have seen, the more readily the next bright idea will come. The next chapter offers a few.

CHAPTER 3

Some Theorems in Number Theory

3.1. The Binomial Theorem

Let n be a positive integer. The **factorial** of n , denoted $n!$, is the product of all positive integers less than or equal to n :

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1.$$

It is easy to see from this definition that for $n \geq 2$ we can write $n! = n \cdot (n-1)!$. For example,

$$5! = 5 \cdot 4! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120.$$

For the formula $n! = n \cdot (n-1)!$ to hold when $n = 1$ as well, we need $1! = 1 \cdot 0!$. Since $1! = 1$, this requires $0! = 1$, and so we define $0!$ to be 1.

Closely connected with the factorial notation is the **binomial coefficient**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

for any positive integer n and any integer k satisfying $0 \leq k \leq n$. For example, with $n = 5$ and $k = 3$, we have

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{5 \cdot 4 \cdot 3!}{3!2!} = \frac{5 \cdot 4}{2!} = \frac{20}{2} = 10.$$

It follows from the definition that

$$\binom{n}{0} = \binom{n}{n} = 1,$$

but it is not immediately obvious from the formula that the binomial coefficient is always a positive integer. To see that it is, we first establish a relation between neighboring binomial coefficients.

Lemma 3.1.1: Pascal's Rule

For any integers n and k with $1 \leq k \leq n$, we have

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

Proof. We bring the two fractions to a common denominator. Since

$$(n - k + 1)! = (n - k + 1) \cdot (n - k)!,$$

and

$$k! = k \cdot (k - 1)!,$$

we have

$$\binom{n}{k} = \frac{n!}{k! (n - k)!} = \frac{n! (n - k + 1)}{k! (n - k + 1)!},$$

and

$$\binom{n}{k - 1} = \frac{n!}{(k - 1)! (n - k + 1)!} = \frac{n! k}{k! (n - k + 1)!}.$$

Upon adding these expressions, we obtain

$$\binom{n}{k} + \binom{n}{k - 1} = \frac{n! (n + 1)}{k! (n + 1 - k)!} = \binom{n + 1}{k}$$

which is what we wanted to prove. $\square$

Proposition 3.1.2

For every positive integer n and every integer k with $0 \leq k \leq n$, the binomial coefficient is a positive integer.

Proof. We use induction on n . When $n = 1$, the only coefficients are

$$\binom{1}{0} = \binom{1}{1} = 1,$$

which are positive integers. Assume that the coefficient is a positive integer for every k with $0 \leq k \leq m$. Consider

$$\binom{m + 1}{k},$$

with $0 \leq k \leq m + 1$. If $k = 0$ or $k = m + 1$, it equals 1. Otherwise $1 \leq k \leq m$, and by Pascal's rule

$$\binom{m + 1}{k} = \binom{m}{k} + \binom{m}{k - 1}.$$

Both k and $k - 1$ lie between 0 and m , so by the induction hypothesis both terms on the right are positive integers, and so is their sum. This completes the induction. $\square$

The binomial theorem is a formula for a complete expansion of the expression $(x + y)^n$ for any positive integer n into a sum of powers

of x and y . It says the coefficient of each term $x^{n-k}y^k$ is precisely the binomial coefficient, or that

$$(x+y)^n = \binom{n}{0} x^n y^0 + \binom{n}{1} x^{n-1} y^1 + \binom{n}{2} x^{n-2} y^2 + \cdots + \binom{n}{n} x^0 y^n.$$

For example,

$$(x+y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5,$$

and here the coefficient of x^2y^3 is 10, which is the binomial coefficient computed above. Using summation notation, the theorem can be stated more concisely as follows.

Theorem 3.1.3: The Binomial Theorem

Let $n \geq 1$ be an integer. Then, for all nonzero real numbers x and y

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

Proof. We prove it by mathematical induction on n . When $n = 1$, the formula becomes

$$(x+y)^1 = \sum_{k=0}^1 \binom{1}{k} x^{1-k} y^k = \binom{1}{0} x^1 y^0 + \binom{1}{1} x^0 y^1 = x + y,$$

which is correct. Assume that the formula holds for some integer $m \geq 1$. We must show that it also must hold for $m+1$. First, notice that

$$(x+y)^{m+1} = (x+y)(x+y)^m = x(x+y)^m + y(x+y)^m.$$

Under the induction hypothesis

$$\begin{aligned} x(x+y)^m &= x \sum_{k=0}^m \binom{m}{k} x^{m-k} y^k \\ &= \sum_{k=0}^m \binom{m}{k} x^{m+1-k} y^k \\ &= \binom{m}{0} x^{m+1} y^0 + \sum_{k=1}^m \binom{m}{k} x^{m+1-k} y^k \\ &= x^{m+1} + \sum_{k=1}^m \binom{m}{k} x^{m+1-k} y^k, \end{aligned}$$

and

$$\begin{aligned}
y(x+y)^m &= y \sum_{j=0}^m \binom{m}{j} x^{m-j} y^j \\
&= \sum_{j=0}^m \binom{m}{j} x^{m-j} y^{j+1} \\
&= \sum_{k=1}^{m+1} \binom{m}{k-1} x^{m+1-k} y^k \\
&= \sum_{k=1}^m \binom{m}{k-1} x^{m+1-k} y^k + \binom{m}{m} x^1 y^{m+1} \\
&= \sum_{k=1}^m \binom{m}{k-1} x^{m+1-k} y^k + y^{m+1}.
\end{aligned}$$

where we used the change of index $k = j+1$, so that $j = k-1$. As j runs from 0 to m , k runs from 1 to $m+1$. Upon adding these expressions and using Pascal's rule, we obtain

$$\begin{aligned}
(x+y)^{m+1} &= x^{m+1} + \sum_{k=1}^m \binom{m}{k} x^{m+1-k} y^k + \sum_{k=1}^m \binom{m}{k-1} x^{m+1-k} y^k + y^{m+1} \\
&= x^{m+1} + \sum_{k=1}^m \left[\binom{m}{k} + \binom{m}{k-1} \right] x^{m+1-k} y^k + y^{m+1} \\
&= x^{m+1} + \sum_{k=1}^m \binom{m+1}{k} x^{m+1-k} y^k + y^{m+1} \\
&= \sum_{k=0}^{m+1} \binom{m+1}{k} x^{m+1-k} y^k
\end{aligned}$$

which is the formula in the case $n = m+1$. □

3.2. The Division Algorithm

The division algorithm describes what happens in a division. If we divide 23 by 4 we get

$$\frac{23}{4} = \frac{20}{4} + \frac{3}{4} = 5 + \frac{3}{4},$$

which says that $23 = 5 \cdot 4 + 3$. We say that 5 is the **quotient** and 3 is the **remainder**. Notice that 23 could also be written as $4 \cdot 4 + 7$ or as $6 \cdot 4 - 1$. What singles out $23 = 5 \cdot 4 + 3$ is that the remainder is neither negative nor as large as 4: we have taken away as many fours as we can,

and what is left is less than one more four. Roughly stated, then, the division algorithm states that an integer a can always be divided by a positive integer b in such a way that the remainder is smaller than b . The exact statement is as follows.

Theorem 3.2.1: The Division Algorithm

Given integers a and b with $b \geq 1$, there exist unique integers q and r satisfying

$$a = qb + r, \quad 0 \leq r < b.$$

Proof. We first show the existence of q and r . Consider the integers of the form $a - xb$ where x is an integer and $a - xb \geq 0$. There is at least one: taking $x = -|a|$ gives

$$a - xb = a + |a|b \geq a + |a| \geq 0,$$

since $b \geq 1$. If 0 is among these integers, it is the smallest, since the others are positive; otherwise they are all positive, and by the well-ordering principle there is a smallest among them. In either case, call the smallest r and let q be the integer with $r = a - qb$. Then $a = qb + r$ and $r \geq 0$. We claim that $r < b$. Assume for the sake of a contradiction that this is not the case. Then $r \geq b$ and

$$0 \leq r - b = a - qb - b = a - (q + 1)b,$$

which is again of the form $a - xb$ and is nonnegative. But $r - b = a - (q + 1)b < r$ since $b > 0$ and this contradicts the choice of r as the smallest. Hence, $r < b$.

We now show uniqueness. Suppose that a has two representations of the desired form, say

$$a = qb + r = q'b + r', \quad 0 \leq r < b, 0 \leq r' < b.$$

Subtracting gives

$$r' - r = b(q - q'),$$

so $r' - r$ is a multiple of b . Since both r and r' lie between 0 and $b - 1$, the largest $r' - r$ can be is

$$(b - 1) - 0 = b - 1,$$

and the smallest it can be is

$$0 - (b - 1) = -(b - 1).$$

Therefore, $-(b - 1) \leq r' - r \leq b - 1$, and in particular $-b < r' - r < b$, which yields

$$-1 < q - q' < 1.$$

Since $q - q'$ is an integer, the only possibility is that $q - q' = 0$, whence $q = q'$. This in turn gives $r' = r$, and the proof is complete. $\square$

Corollary 3.2.2

Let a and b be integers with $b \neq 0$. Then there exist unique integers q and r such that

$$a = qb + r, \quad 0 \leq r < |b|.$$

Proof. If $b > 0$, then $|b| = b$ and this is Theorem 3.2.1. So suppose $b < 0$. Then $|b| > 0$, and Theorem 3.2.1 gives unique integers q' and r such that

$$a = q'|b| + r, \quad 0 \leq r < |b|.$$

Since $|b| = -b$, we may take $q = -q'$ to arrive at $a = qb + r$ with $0 \leq r < |b|$, and uniqueness of q and r follows from that of q' and r . $\square$

The division algorithm does more than describe a division. Taking $b = 2$, it says that every integer can be written in exactly one of the forms $2k$ and $2k + 1$; that is, every integer is either even or odd, and none is both. Nor is there anything special about 2. Taking $b = 3$, every integer has exactly one of the forms $3k$, $3k + 1$ and $3k + 2$; taking $b = 4$, exactly one of four forms; and so on. A claim about infinitely many integers can thus be reduced to a claim about finitely many forms, each of which can be settled by ordinary arithmetic.

Proposition 3.2.3

The square of every integer has the form $4k$ or $4k + 1$ for some integer k .

Proof. Let n be an integer. By the division algorithm with $b = 2$, either $n = 2q$ or $n = 2q + 1$ for some integer q . If $n = 2q$, then

$$n^2 = (2q)^2 = 4q^2,$$

which has the form $4k$ with $k = q^2$. If $n = 2q + 1$, then

$$n^2 = (2q + 1)^2 = 4q^2 + 4q + 1 = 4(q^2 + q) + 1,$$

which has the form $4k + 1$ with $k = q^2 + q$. There is no other case, so the proposition holds for every integer. $\square$

Corollary 3.2.4

No integer of the form $4k + 3$ is a sum of two squares.

Proof. Let m and n be integers. By Proposition 3.2.3 $m^2 = 4j$ or $m^2 = 4j + 1$, and $n^2 = 4q$ or $4q + 1$ for some integers j and q . Adding, we find that $m^2 + n^2$ is one of

$$4(j + q), \quad 4(j + q) + 1, \quad 4(j + q) + 2,$$

so its remainder on division by 4 is 0, 1 or 2. By the uniqueness of the division algorithm, an integer has only one remainder on division by 4, so $m^2 + n^2$ cannot have a remainder 3. That is, it is not of the form $4k + 3$. $\square$

3.3. The Fundamental Theorem of Arithmetic

In the preamble we stated, without proof, that every integer greater than 1 can be written as a product of primes. In this section we prove it, and prove more: that the product is unique, apart from the order in which the primes are written. Thus $60 = 2 \cdot 2 \cdot 3 \cdot 5$, and however we go about factoring 60, we shall always arrive at two 2s, one 3 and one 5. The crucial result we need to prove it is called **Euclid's lemma**.

Lemma 3.3.1: Euclid's Lemma

Let p be a prime and let a and b be integers. If $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. If $p \mid a$, there is nothing to prove, so assume $p \nmid a$; we shall show that $p \mid b$. Note that $a \neq 0$ since p divides 0. Consider all integers $x > 0$ such that $p \mid xb$. We can take $x = p$ since $p \mid pb$. We can also take $x = |a|$ since $|a|b$ is either ab or $-ab$ and $p \mid ab$. By the well-ordering principle, there is a smallest such integer; call it d .

We first show that if $p \mid xb$, then $d \mid x$. By the division algorithm, $x = qd + r$ with $0 \leq r < d$. Then,

$$rb = xb - q(db).$$

Since p divides both xb and db , say $xb = ps$ and $db = pt$, we have $rb = p(s - qt)$, so $p \mid rb$. Now, if r were positive, it would be a positive integer smaller than d with $p \mid rb$, contradicting the choice of d as the smallest such integer. Hence, $r = 0$ and $d \mid x$.

In particular, $d \mid p$. Since p is prime and d is positive, $d = 1$ or $d = p$. But d also divides $|a|$ and if d were p , then p would divide a

contrary to our assumption. Hence, $d = 1$, which means that $p \mid 1 \cdot b$, that is, $p \mid b$. $\square$

The lemma extends at once from two factors to any number of them.

Corollary 3.3.2

Let p be prime. If $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i . In particular, if $q_1, \dots, q_n$ are primes and $p \mid q_1 q_2 \cdots q_n$, then $p = q_i$ for some i .

Proof. We use induction on n . When $n = 1$ there is nothing to prove. Assume that the statement holds for products of m factors, and suppose that $p \mid a_1 \cdots a_m a_{m+1}$. Writing this product as $(a_1 \cdots a_m) a_{m+1}$, Euclid's lemma gives $p \mid a_1 \cdots a_m$ or $p \mid a_{m+1}$. In the first case, the induction hypothesis gives $p \mid a_i$ for some $i \leq m$. In the second, $p \mid a_{m+1}$. In either case, p divides one of the factors.

For the second statement, if $p \mid q_1 q_2 \cdots q_n$ then we have just shown that $p \mid q_i$ for some i . Being a prime, q_i is not divisible by any positive integer other than 1 and q_i itself. Because $p > 1$, we are forced to conclude that $p = q_i$. $\square$

We can now prove the theorem.

Theorem 3.3.3: The Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written as a product of primes, and this product is unique apart from the order of the factors.

Proof. Suppose, for a contradiction, that there is at least one integer $n \geq 2$ that cannot be written as a product of primes. Among all such integers, let m be the smallest. Then m is not prime, for a prime is a product of one prime. So m is composite. Hence, $m = ab$ for some integers a and b with $2 \leq a, b < m$. Since a and b are greater than 1 and smaller than m , each is a product of primes, by the choice of m . But then so is $m = ab$, which contradicts the choice of m .

It remains to prove uniqueness, and we argue as we did for existence. Suppose, for a contradiction, that some integer greater than 1 has two factorizations into primes that differ other than in the order of the factors. Among all such integers, let n be the smallest, and write

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

Since p_1 divides $n = q_1 q_2 \cdots q_s$, Corollary 3.3.2 gives $p_1 = q_j$ for some j . Renumbering the q 's, we take $j = 1$. If $r = 1$, then $n = p_1 = q_1$, and

the remaining primes $q_2, \dots, q_s$, if there were any, would have a product equal to 1. This is impossible, since every prime is greater than 1. So $s = 1$ as well, and the two factorizations are the same, contrary to our assumption. The same argument applies if $s = 1$.

Otherwise, dividing both factorizations by $p_1 = q_1$ gives

$$m = \frac{n}{p_1} = p_2 \cdots p_r = q_2 \cdots q_s,$$

where $1 < m < n$. These two factorizations of m cannot differ only in order. If they did, then restoring the common prime $p_1 = q_1$ to both would show that the two factorizations of n differ only in order. So m is a smaller integer with two genuinely different factorizations, contradicting the choice of n . Hence, the factorization is unique. $\square$

3.4. Fermat's Little Theorem

Take any prime, say 5, and any positive integer, say 2. Raise the integer to the power of the prime and subtract the integer itself

$$2^5 - 2 = 30,$$

which is divisible by 5. Try 3 instead:

$$3^5 - 3 = 240,$$

again divisible by 5. With the prime 7:

$$2^7 - 2 = 126 = 7 \cdot 18.$$

The pattern is not an accident. In a letter of 1640 to Frénicle de Bessy, Pierre de Fermat stated, in substance, that if p is a prime and a an integer, then $a^p - a$ is divisible by p . He gave no proof. Leibniz found one some decades later but did not publish it, and the first presented proof, by Euler, appeared in 1736. The result is known as **Fermat's little theorem**, to distinguish it from a more famous statement of his that took three and a half centuries to prove.

Lemma 3.4.1

Let p be a prime and k an integer with $0 < k < p$. Then

$$p \mid \binom{p}{k}$$

Proof. By Proposition 3.1.2, the binomial coefficient is an integer, and by definition of the binomial coefficient,

$$p! = \binom{p}{k} \cdot k! (p-k)!,$$

and p divides $p!$. By Euclid's lemma, then p divides $\binom{p}{k}$ or p divides $k! (p-k)!$. We show that the second is impossible. The product $k! (p-k)!$ is a product of positive integers, each at most k or at most $p-k$, and so each smaller than p . A prime cannot divide a positive integer smaller than itself, so p divides none of these factors, and by Corollary 3.3.2 it does not divide their product. Hence, $p \mid \binom{p}{k}$. $\square$

Theorem 3.4.2: Fermat's Little Theorem

Let p be prime. Then $p \mid a^p - a$ for every positive integer a .

Proof. We use induction on a . When $a = 1$, we have $1^p - 1 = 0$, which is divisible by p . Assume that $p \mid m^p - m$ for some positive integer m . By the binomial theorem with $x = 1$ and $y = m$,

$$(m+1)^p = \sum_{k=0}^p \binom{p}{k} m^k = 1 + \sum_{k=1}^{p-1} \binom{p}{k} m^k + m^p,$$

where we have separated the first and last terms. Subtracting $m+1$ from both sides and rearranging,

$$(m+1)^p - (m+1) = (m^p - m) + \sum_{k=1}^{p-1} \binom{p}{k} m^k.$$

The first term on the right is divisible by p by the induction hypothesis, and each term of the sum is divisible by p by Lemma 3.4.1. Hence p divides the whole right-hand side, and so

$$p \mid (m+1)^p - (m+1).$$

This completes the induction. $\square$

The theorem is often stated in a second form, which follows from the first by one more application of Euclid's lemma.

Corollary 3.4.3

Let p be prime and a a positive integer with $p \nmid a$. Then, $p \mid a^{p-1} - 1$

Proof. By Fermat's Little Theorem, p divides $a^p - a = a(a^{p-1} - 1)$. Since $p \nmid a$, Euclid's lemma gives $p \mid a^{p-1} - 1$. $\square$

Consider what went into the proof of Fermat's little theorem. Induction carried the claim from each integer to the next. The binomial theorem, itself proved by induction, supplied the expansion. Euclid's lemma, proved by the well-ordering principle and the division algorithm, decided which terms were divisible by p . None of these results says anything about $a^p - a$; each was proved for its own sake. Yet together they settle a question that none of them could settle alone. This is how mathematics grows: results proved one at a time become the steps of later proofs, and the more of them we have, the further we can reach. Fermat was sure of his theorem, and he was right. He had checked it, no doubt, in many cases, and he may well have had a proof in mind. But until a proof was written down, his certainty was his own, and could not be passed on. Euler's proof could. Anyone who follows it is in the same position as Euler: not persuaded, but compelled.

I now leave you to venture out and discover many more proofs on your own. The road goes ever on; may yours be a good one.

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